



UNIVERSITAT DE  
BARCELONA

Facultat de Física

Degree on Physics

Final Degree Project

# Steady-State Bifurcation in Nonlinear Opinion Dynamics: Analysis on Canonical Networks

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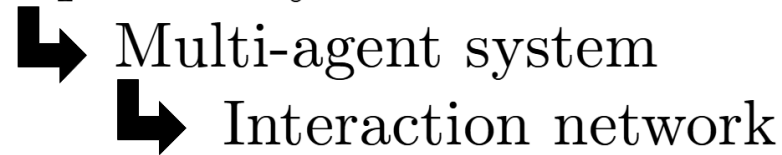
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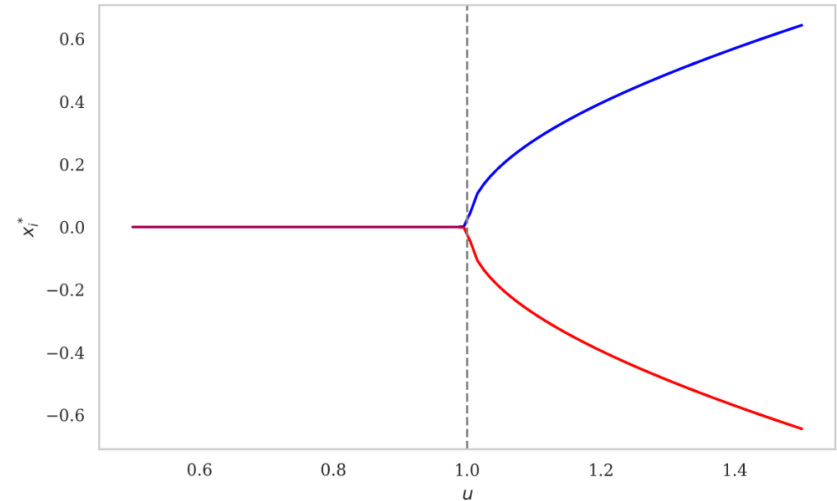
# 1. Objectives and Introduction

Opinion dynamics model



## Objectives:

- Explore steady-state bifurcation in a nonlinear opinion dynamics model
- Apply this framework to canonical networks: regular, star, Watts-Strogatz, and scale-free



## 2. Nonlinear Opinion Dynamics Model

Network of  $N$  agents

Interaction matrix  $\mathbf{\Gamma} = (\gamma_{ij})$

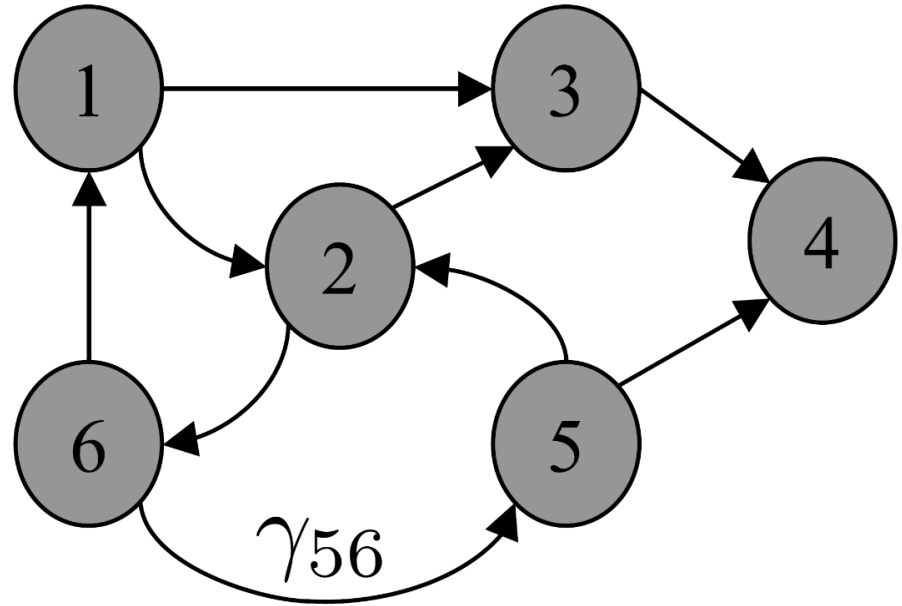
Agent's opinion  $x_i$

- $x_i > 0$  : Favour state
- $x_i < 0$  : Disfavour state
- $x_i = 0$  : Neutral state

Opinion intensity  $|x_i|$

Network opinion vector

$$\mathbf{x} = (x_1, \dots, x_N)$$



## 2. Nonlinear Opinion Dynamics Model

$$\frac{dx_i}{dt} = -dx_i + u \tanh \left( \alpha x_i + \sum_{j \neq i} \gamma_{ij} x_j \right) + b_i$$

Damping parameter  $d$

Attention parameter  $u$

Self-reinforcing parameter  $\alpha$

Bias (or input) parameter  $b_i$

Network bias vector

$\mathbf{b} = (b_1, \dots, b_N)$

### 3. Unbiased Opinion Dynamics

$$\frac{dx_i}{dt} = -dx_i + u \tanh(\alpha x_i + \sum_{j \neq i} \gamma_{ij} x_j) + \cancel{b_i}$$

$$\mathbf{b} = \mathbf{0} \quad \Rightarrow \quad \text{Neutral state } \mathbf{x} = \mathbf{0} \text{ is stationary}$$

Let us examine the stability of the neutral state

### 3. Unbiased Opinion Dynamics

#### Bifurcation Theorem:

Suppose  $\mathbf{\Gamma} = \gamma \mathbf{A}$ , where  $\gamma > 0$  and  $\mathbf{A}$  is a **Perron-Frobenius** matrix:

- Non-negative
- Irreducible
- Has positive largest e-val  $\lambda_{PF}$ , with associated e-vec  $\mathbf{v}^* = \mathbf{v}_{PF} \in (\mathbb{R}^+)^N$

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Let:

Then:

$$u^* = \frac{d}{\alpha + \gamma \lambda_{PF}}$$

- a) If  $u < u^*$ , then  $\mathbf{x} = \mathbf{0}$  is locally stable
- b) If  $u > u^*$ , then  $\mathbf{x} = \mathbf{0}$  is locally unstable

Threshold parameter

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Let:

Then:

$$u^* = \frac{d}{\alpha + \gamma \lambda_{PF}}$$

c) For  $u = u^*$ , **steady-state bifurcation**  
at  $(\mathbf{x}, u) = (\mathbf{0}, u^*)$ :

Threshold parameter

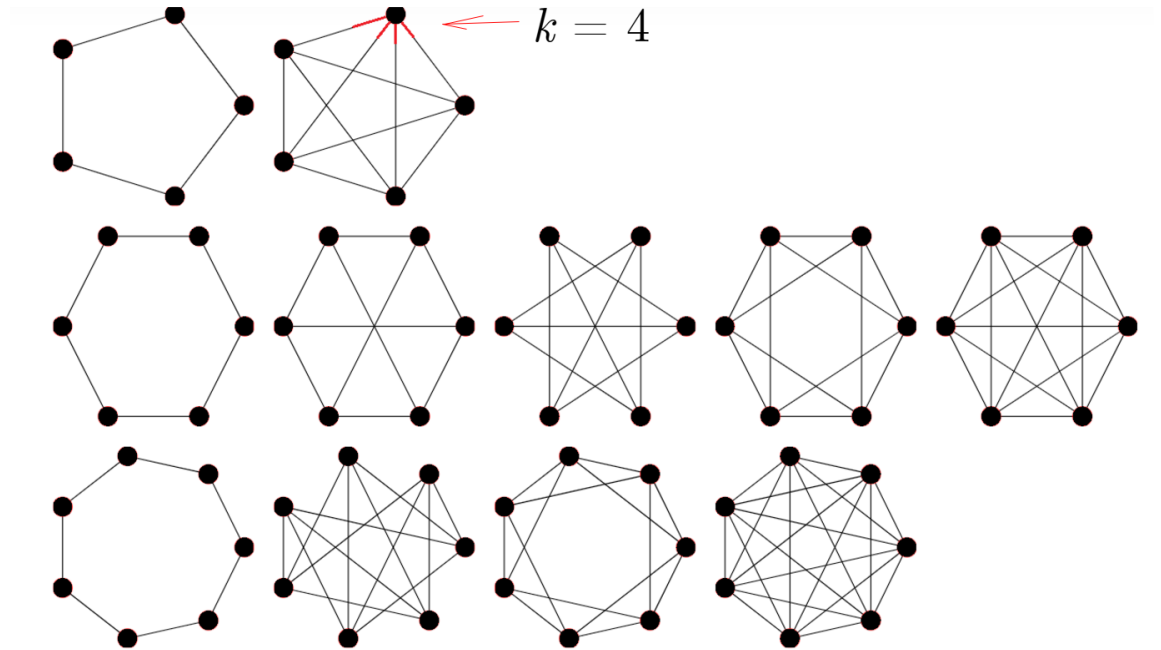
$$\mathbf{x}^*(u \gtrsim u^*) \approx r \mathbf{v}^* \quad \text{Agreement states}$$

## 4. Application to Particular Graphs

### A. Regular Graphs

$$u^* = \frac{d}{\alpha + \gamma k}$$

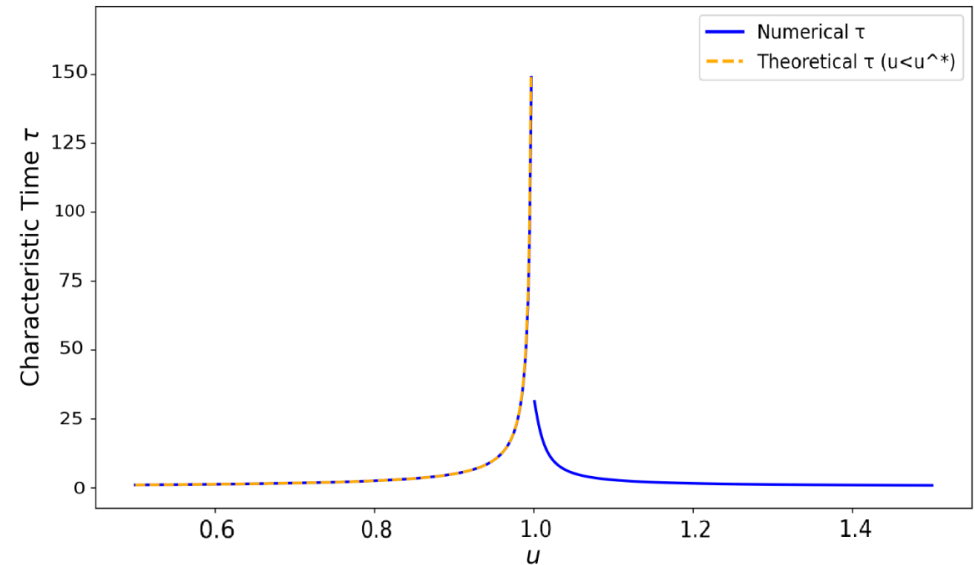
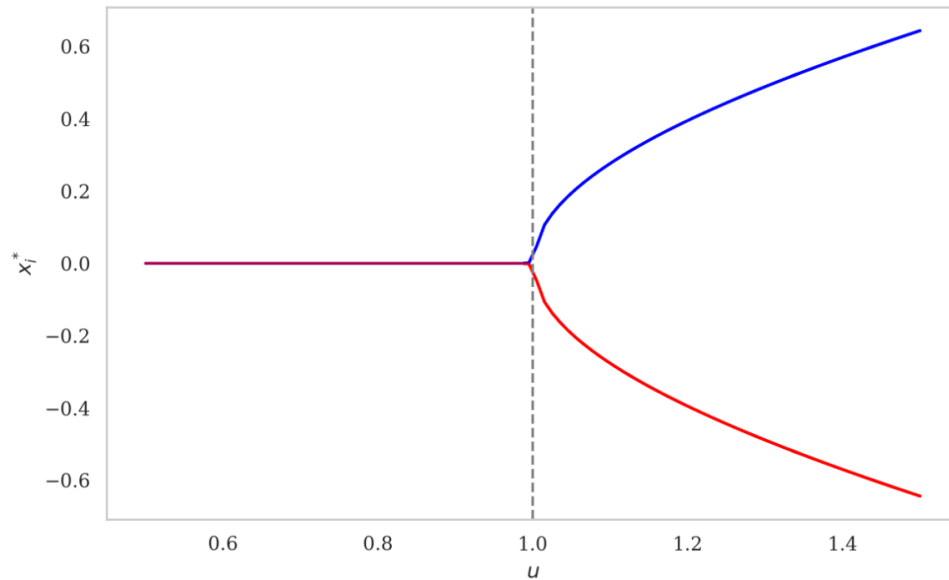
$$\mathbf{v}^* = \frac{1}{\sqrt{N}}(1, \dots, 1)$$



## 4. Application to Particular Graphs

$$\gamma = 0.1 \quad \alpha = 1 \quad d = 2$$

### A. Regular Graphs



$$N = 100 \quad k = 10 \quad u^* = 1$$

$$\tau = \frac{1}{d(1-u/u^*)}$$

## 4. Application to Particular Graphs

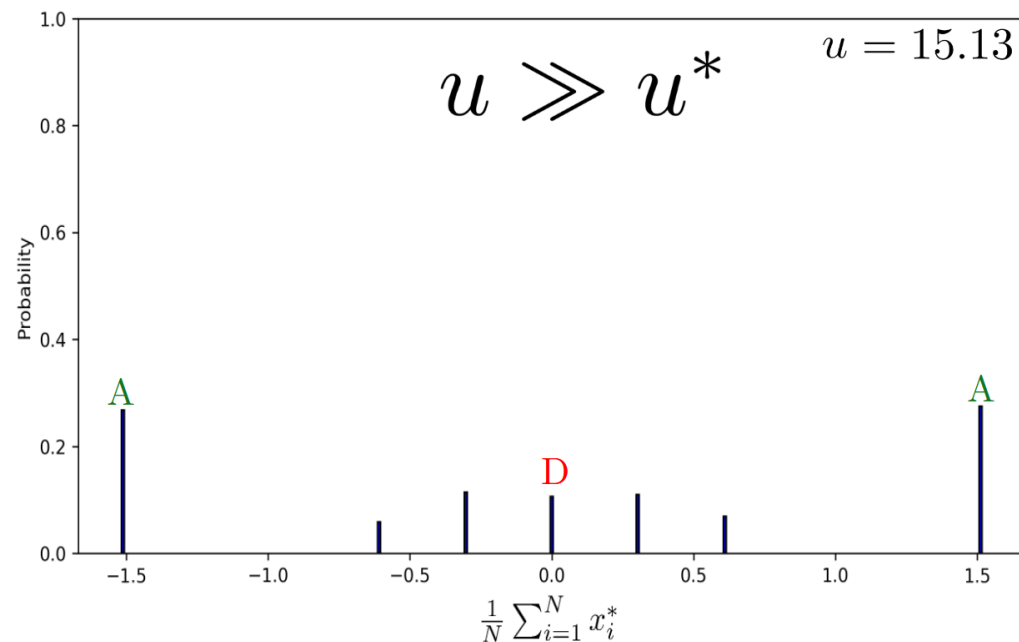
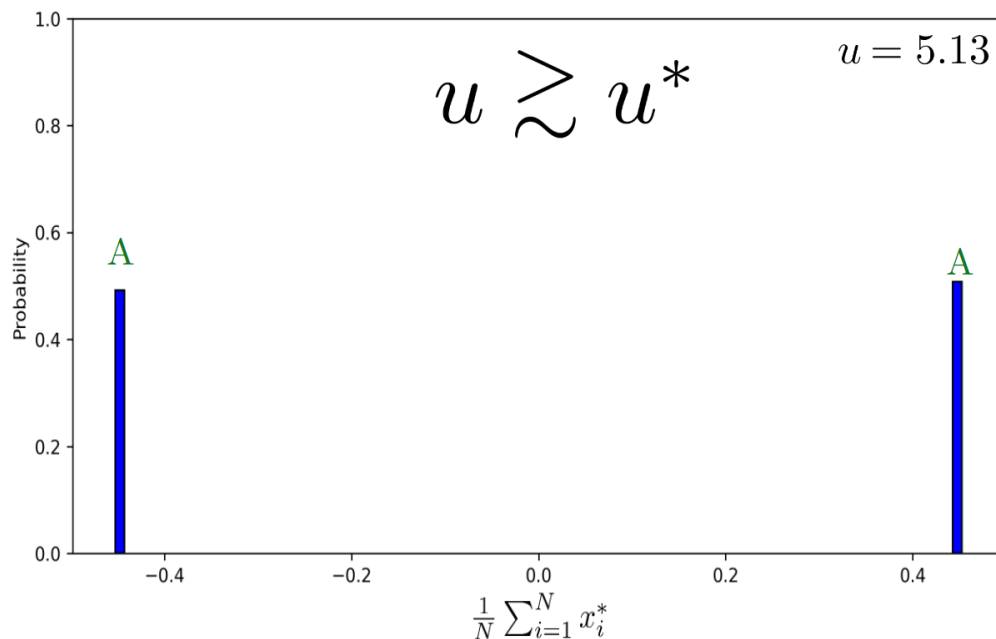
$$\gamma = 1.5 \quad \alpha = 0 \quad d = 10$$

$$N = 10, k = 2$$

$$u^* = 3.33$$

1,000 ICs

### A. Regular Graphs: Solutions beyond the threshold...



A: Agreement

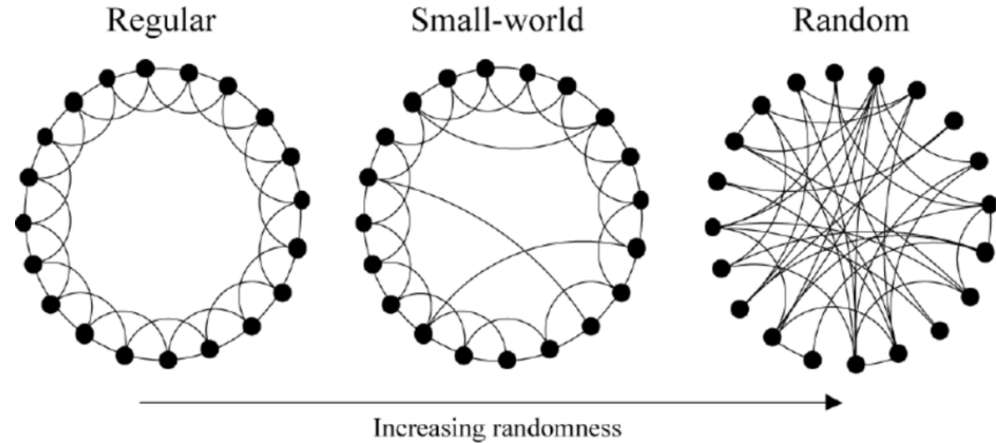
D: Disagreement

## 4. Application to Particular Graphs

### B. Watts-Strogatz Graphs

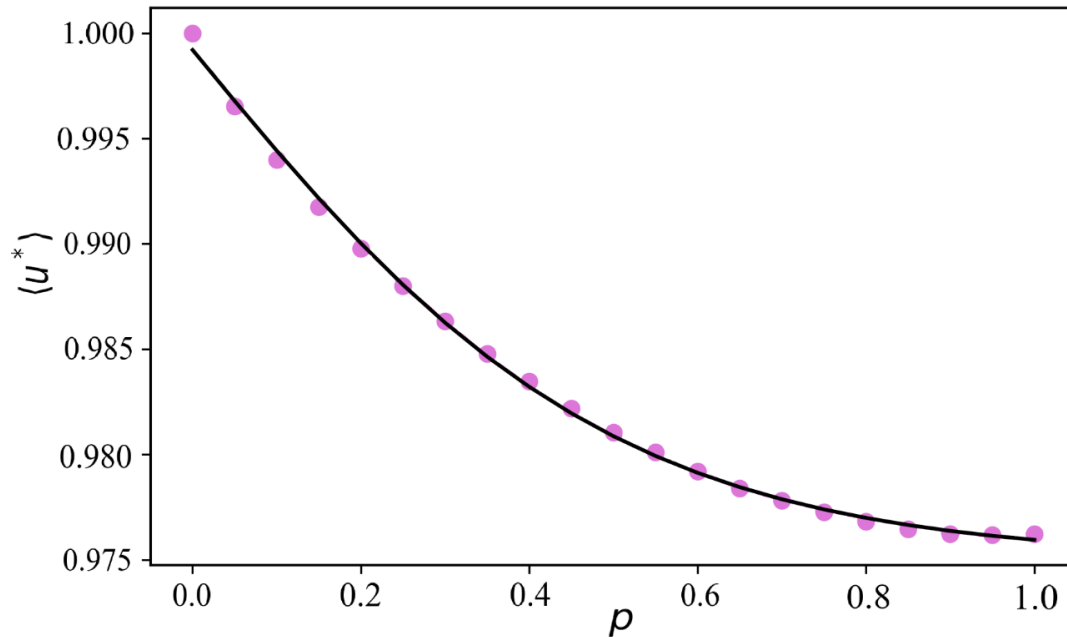
$$\langle u^* \rangle(p, N) \quad ?$$

$$\langle \mathbf{v}^* \rangle = \frac{1}{\sqrt{N}}(1, \dots, 1)$$



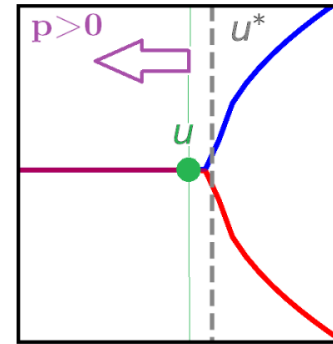
## 4. Application to Particular Graphs

### B. Watts-Strogatz Graphs

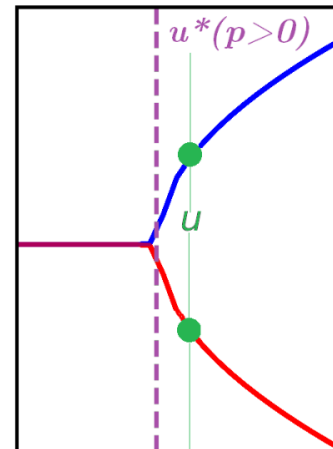


$N = 100$      $k = 10$     10,000 repetitions

$$\gamma = 0.1 \quad \alpha = 1 \quad d = 2$$



Randomness



## 5. Opinion Unfolding in the Presence of Input

$$\frac{dx_i}{dt} = -dx_i + u \tanh(\alpha x_i + \sum_{j \neq i} \gamma_{ij} x_j) + \textcircled{b_i}$$

$\mathbf{b} \neq \mathbf{0} \quad \Rightarrow \quad \text{Neutral state } \mathbf{x} = \mathbf{0} \text{ is NOT stationary}$

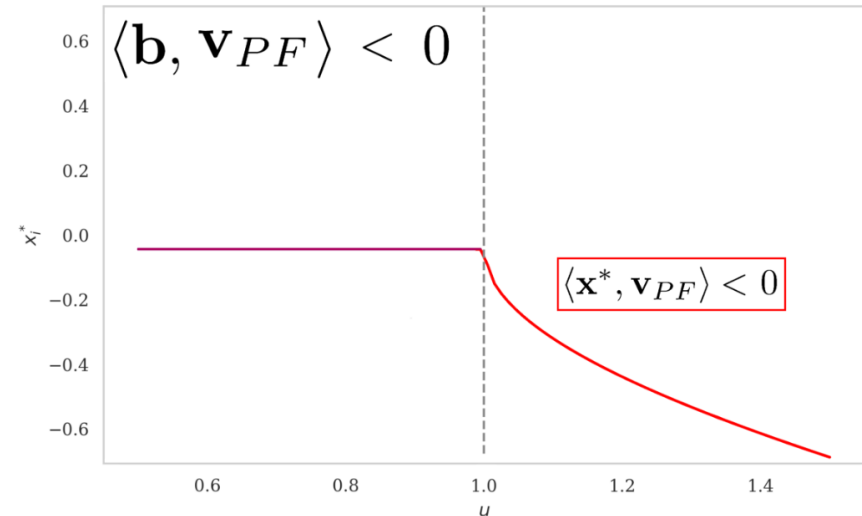
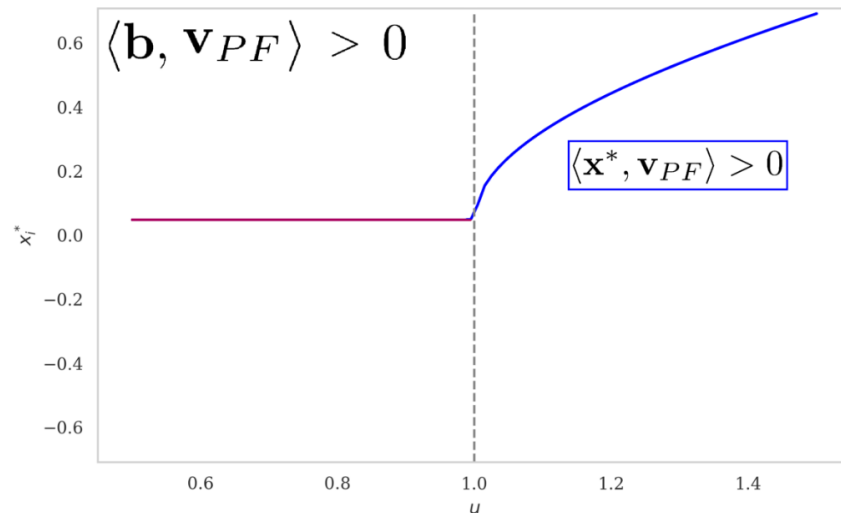
How does the bifurcation evolve?

## 5. Opinion Unfolding in the Presence of Input

### Bifurcation Unfolding:

If  $\Gamma = \gamma \mathbf{A}$  (symmetric):

- a) If  $\langle \mathbf{b}, \mathbf{v}_{PF} \rangle = 0$ , then the bifurcation of opinions remains
- b) If  $\langle \mathbf{b}, \mathbf{v}_{PF} \rangle \neq 0$ , then only one of the symmetric equilibrium branches corresponds to stationary solutions:



## 6. Conclusions

- 1) Emergence of a steady-state bifurcation of opinions at a threshold value of the attention parameter
- 2) Bifurcation branches are proportional to the eigenvector associated with the largest eigenvalue of the network
- 3) The characteristic time diverges at the threshold
- 4) In a Watts-Strogatz graph, the average threshold parameter decreases as the rewiring probability increases
- 5) In the presence of bias, the bifurcation unfolds towards one of the symmetric stationary states

# References

- [1] A. Bizyaeva, A. Franci, N. E. Leonard, *Nonlinear Opinion Dynamics with Tunable Sensitivity*, arXiv:2009.04332v3 [math.OC], (2021).
- [2] A. Bizyaeva, T. Sorochnik, A. Franci, N. E. Leonard, *Control of agreement and disagreement cascades with distributed inputs*, arXiv:2103.14764, (2021).
- [3] M. Golubitsky, I. Stewart, *The Symmetry Perspective: From Equilibrium to Chaos in Phase Space and Physical Space*, Progress in Mathematics, **200**, Birkhäuser Verlag, (2002).
- [4] P. Van Mieghem, *Graph Spectra for Complex Networks*, Cambridge University Press, (2011).
- [5] C. Sarkar, S. Jalan, *Spectral properties of complex networks*, arXiv preprint arXiv:1810.01254, (2018).
- [6] F. R. K. Chung, *Spectral Graph Theory*, CBMS Regional Conference Series in Mathematics, **92**, American Mathematical Society, (1996).
- [7] F. R. Gantmacher, *The Theory of Matrices*, Vol. 2, Chelsea Publishing Company, (1959).